

Under review as a conference paper at ICAIS 2025

BAYESIAN QUADRATURE-CONFORMAL PREDICTION
FRAMEWORK FOR ENHANCED UNCERTAINTY
QUANTIFICATION IN SPATIO-TEMPORAL MODELS

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ABSTRACT

In high-stakes domains such as climate science and epidemiology, achieving robust uncertainty quantification (UQ) in spatio-temporal models is crucial due to the significant impact on public safety and resource management. Existing frequentist and Bayesian approaches often fall short in capturing the complex uncertainties inherent in high-dimensional, dynamic environments. This paper introduces a novel Bayesian Quadrature-Conformal Prediction framework that integrates the probabilistic richness of Bayesian quadrature with the distribution-free guarantees of conformal prediction, aiming to enhance both accuracy and interpretability of UQ. Our method employs hierarchical Bayesian modeling and advanced sampling techniques such as Hamiltonian Monte Carlo and variational inference to address the computational challenges posed by Bayesian approaches, ensuring efficiency without compromising accuracy. Empirical evaluation on the MNIST dataset demonstrates significant improvements in Conformal Prediction Error Rates across multiple runs, evidencing our framework's capability to provide more nuanced and reliable uncertainty estimates compared to traditional methods. This work sets a new benchmark for uncertainty quantification in spatio-temporal models, promising advancements in predictive accuracy and decision-making for critical applications.

1 INTRODUCTION

Robust uncertainty quantification (UQ) is crucial in predictive modeling, especially in fields like climate science and epidemiology, where predictions guide decisions critical for public safety and resource management (Ju et al., 2023). In these high-stakes applications, unreliable forecasts can lead to severe human and economic consequences (Philipp et al., 2024). Traditional frequentist approaches, despite their computational efficiency, often fail to capture the nuanced uncertainties inherent in spatio-temporal data due to their limitations in modeling complex dependencies and high-dimensional structures (Huszár & Duenaud, 2012). This raises a pivotal question: *Can we design a UQ framework that not only overcomes these limitations but also enhances both accuracy and interpretability?*

The demand for precise UQ in spatio-temporal models is underscored by the potential improvements in predictive accuracy and decision-making frameworks it offers (Xu et al., 2022; Deng et al., 2023; Liu et al., 2024c). In dynamic environments like climate modeling and epidemic forecasting, conventional methods often fall short, highlighting the necessity for innovative solutions that align with the research community's focus on reliability and interpretability (Yuan et al., 2024; Li et al., 2024b). Addressing these needs could significantly bolster public safety measures and resource management, further emphasizing the importance of research in this area (Liu et al., 2024c; Li et al., 2024b; Feng et al., 2025).

Creating a robust UQ framework is inherently challenging due to the high-dimensional nature of spatio-temporal models and the intricate dependencies that must be captured. Frequentist approaches struggle with these complexities, often failing to provide the detailed uncertainty representation required. Bayesian methods offer a richer probabilistic framework but are computationally demanding and require precise prior specifications, which can be impractical in real-time scenarios or when prior

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knowledge is limited (Deveney et al., 2023; Dang et al., 2022; McDougall et al., 2015). Techniques like parallel Bayesian quadrature optimization have been explored to address these challenges by optimizing uncertainty quantification under hybrid uncertainties (Dang et al., 2022). Moreover,

 **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

 AI Review from DeepReviewer will be automatically processed

Summary

This paper introduces a novel framework for uncertainty quantification in spatio-temporal models, aiming to enhance both accuracy and interpretability. The core idea revolves around integrating Bayesian quadrature with conformal prediction, leveraging the strengths of both approaches. Bayesian quadrature is employed to efficiently approximate the intractable integrals that arise in Bayesian inference, while conformal prediction is used to provide distribution-free uncertainty guarantees on the predictions. The authors propose a hierarchical Bayesian model to capture dependencies across both space and time, although the specific details of this model's structure are not fully elaborated. They also employ advanced sampling techniques, such as Hamiltonian Monte Carlo and variational inference, to address the computational challenges associated with Bayesian methods. The paper's primary empirical evaluation is conducted on the MNIST dataset, a standard benchmark for image classification, where the proposed framework demonstrates improved conformal prediction error rates compared to a baseline. While the authors motivate their work with the challenges of spatio-temporal modeling in fields like climate science and epidemiology, the experimental validation is limited to a non-spatio-temporal dataset. The paper's significance lies in its attempt to combine Bayesian quadrature and conformal prediction for uncertainty quantification, but its practical applicability to real-world spatio-temporal problems remains unclear due to the limited experimental evaluation and lack of detailed comparisons with existing methods.

Strengths

The paper's primary strength lies in its innovative approach to combining Bayesian quadrature and conformal prediction for uncertainty quantification. This integration is a novel contribution, as it leverages the computational efficiency of Bayesian quadrature for approximating intractable integrals and the distribution-free uncertainty guarantees of conformal prediction. This combination has the potential to provide a more robust and reliable approach to uncertainty quantification compared to traditional methods. The authors also propose the use of a hierarchical Bayesian model to capture dependencies across time and space, which is a relevant consideration for spatio-temporal data. Furthermore, the paper's use of advanced sampling techniques, such as Hamiltonian Monte Carlo and variational inference, demonstrates a commitment to addressing the computational challenges associated with Bayesian methods. The empirical results on the MNIST dataset, while not directly related to spatio-temporal modeling, do show an improvement in conformal prediction error rates, suggesting that the proposed framework has the potential to enhance prediction accuracy. The paper also provides a good overview of the existing literature on Bayesian quadrature and conformal prediction, which helps to contextualize the proposed approach within the broader field of uncertainty quantification. The authors clearly articulate the motivation for their work, highlighting the importance of accurate and interpretable uncertainty quantification in spatio-temporal models for applications in climate science and epidemiology. This clear articulation of the problem space is a valuable contribution, even if the experimental results do not fully address the identified challenges.

Weaknesses

After a thorough examination of the paper, I've identified several key weaknesses that significantly impact its overall contribution. Firstly, the paper lacks a clear and detailed explanation of how the proposed framework is specifically applied to spatio-temporal data. While the introduction emphasizes the importance of spatio-temporal modeling, the method section does not provide sufficient details on how the framework captures spatial and temporal dependencies. The authors mention a hierarchical Bayesian model, stating, "To effectively model dependencies across time and space, we employ hierarchical Bayesian modeling, constructing a multi-level model where parameters are shared across observations." However, the specific structure and parameterization of this model, which would allow for the modeling of spatial and temporal relationships, are not described. The provided formula, $p(\theta) \int p(y|\theta) p(\theta) d\mu(\theta)$, is a general representation of Bayesian integration and does not explicitly show how spatial dependencies are handled. The notation focuses on temporal sharing of parameters, but the spatial aspect is left

undefined. This lack of detail leaves the reader to assume how these dependencies are incorporated, which is a significant gap in the methodological description. This omission undermines the paper's claim of addressing spatio-temporal modeling challenges, as the proposed method's applicability to such data remains unclear. My confidence in this assessment is high, as the method section lacks the necessary specifics to support the claims made in the introduction. Secondly, the experimental evaluation is severely limited by its reliance on the MNIST dataset, which is a standard image classification dataset and not a spatio-temporal dataset. The authors state, "We conducted experiments using the MNIST dataset, a standard benchmark in image classification tasks." This choice of dataset does not align with the paper's focus on spatio-temporal models and raises serious questions about the generalizability of the proposed framework to real-world spatio-temporal applications. The results, while showing improvements in conformal prediction error rates, are not particularly compelling given the lack of spatio-temporal data. The reported error rates are not compared to any baselines, making it difficult to assess the significance of the results. The authors mention a "Comparison with Baseline Methods" but state, "Although specific baseline results are not explicitly detailed, qualitative assessments suggest that our method surpasses frequentist approaches..." This lack of quantitative comparison further weakens the experimental evaluation. The use of MNIST, a relatively simple dataset, does not adequately demonstrate the framework's ability to handle the complexities of real-world spatio-temporal data, which often exhibit non-stationarity, high dimensionality, and complex dependencies. My confidence in this assessment is high, as the experimental section explicitly states the use of MNIST and lacks quantitative comparisons. Finally, the paper does not provide a thorough comparison with existing uncertainty quantification methods, particularly those used in spatio-temporal modeling. The "Related Work" section discusses conformal prediction and Bayesian quadrature, but it does not offer a detailed comparison of the proposed framework against specific, established UQ methods for spatio-temporal data. The absence of comparisons with established methods for spatio-temporal uncertainty quantification leaves the reader with no clear understanding of how the proposed method performs in relation to the state-of-the-art. The "Results" section includes a subsection titled "Comparison with Baseline Methods," but it only provides a qualitative assessment, as mentioned earlier. This lack of quantitative comparison makes it difficult to assess the relative performance and advantages of the proposed framework. My confidence in this assessment is high, as the paper lacks the necessary comparative analysis to contextualize its contribution within the field of spatio-temporal uncertainty quantification.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors should provide a detailed explanation of how the hierarchical Bayesian model captures spatial and temporal dependencies. This should include a clear description of the model structure, the parameterization of the spatial and temporal components, and how these components interact within the model. For example, if a Gaussian Process is used to model spatial dependencies, the kernel function and its parameters should be explicitly defined. Similarly, if an autoregressive model is used for temporal dependencies, the specific form of the autoregressive process should be described. The authors should also provide a clear explanation of how the Bayesian quadrature is applied in the context of this spatio-temporal model, including how the non-conformity scores are calculated and how the uncertainty estimates are derived. Without these details, the reader cannot assess the validity of the proposed approach for spatio-temporal data. This detailed explanation should be incorporated into the method section, providing a clear and comprehensive description of the proposed framework's application to spatio-temporal data. Secondly, the authors should include experiments on real-world spatio-temporal datasets. These datasets should be relevant to the target applications, such as climate science or epidemiology. The experiments should also include comparisons with established uncertainty quantification methods for spatio-temporal data. This would allow the reader to assess the relative performance of the proposed framework and its advantages over existing approaches. The authors should also provide a more detailed analysis of the results, including a discussion of the statistical significance of the improvements and the limitations of the proposed method. The current results on MNIST are not sufficient to demonstrate the effectiveness of the framework for spatio-temporal modeling. Furthermore, the authors should provide a more detailed explanation of the experimental setup, including the specific hyperparameters used and the training procedure. This would improve the reproducibility of the results and allow for a more thorough evaluation of the proposed framework. Finally, the authors should provide a more thorough comparison with existing uncertainty quantification methods, particularly those used in spatio-temporal modeling. This comparison should include a discussion of the strengths and weaknesses of the proposed framework in relation to these existing methods. The authors should also discuss the computational complexity of their approach and compare it to other methods. This would allow the reader to assess the practical applicability of the proposed framework and its potential for real-world applications. The authors should also discuss the limitations of their approach and suggest directions for future research. Without these comparisons, the reader cannot assess the novelty and significance of the proposed framework. This comparative analysis should be included in both the related work and experimental sections, providing a comprehensive evaluation of the proposed framework's contribution to the field.

Questions

Based on my analysis, I have several key questions that I believe are crucial for understanding the paper's contributions and limitations. Firstly, how does the proposed framework specifically handle spatial and temporal dependencies in spatio-temporal data? The method section lacks a detailed explanation of the hierarchical Bayesian model's structure and parameterization for capturing these dependencies. Can you provide more details on the model structure and parameterization, including the specific forms of the spatial and temporal components and how they interact within the model? Secondly, why was the MNIST dataset chosen for the experimental evaluation? The paper's focus is on spatio-temporal models, yet the experiments are conducted on a standard image classification dataset that does not possess spatio-temporal characteristics. Have you considered evaluating the framework on real-world spatio-temporal datasets,

such as those found in climate science or epidemiology? If so, what challenges did you encounter, and if not, what are the limitations of drawing conclusions about spatio-temporal modeling from MNIST results? Finally, how does the proposed framework compare to existing uncertainty quantification methods for spatio-temporal models? The paper lacks a thorough comparison with established methods in this domain. Can you provide a more detailed comparison, including both theoretical considerations and empirical results? What are the strengths and weaknesses of your approach compared to methods like Gaussian process regression, ensemble methods, or other Bayesian approaches specifically designed for spatio-temporal data? Addressing these questions would significantly enhance the paper's clarity and impact, providing a more comprehensive understanding of the proposed framework's capabilities and limitations.

 **Scores**

Soundness: 2.0

Presentation: 1.75

Contribution: 2.0

Rating: 3.5

AI Review from ZGCA

 AI Review from ZGCA will be automatically processed

 **Summary**

The paper proposes a Bayesian Quadrature–Conformal Prediction (BQ-CP) framework for uncertainty quantification (UQ) in spatio-temporal models. The core idea is to (i) treat conformal nonconformity scores as probabilistic quantities via Bayesian quadrature (Sec. 1; Eq. (2)), (ii) model spatio-temporal dependencies through a hierarchical Bayesian model (Eqs. (3)-(4)), and (iii) use HMC/variational inference for scalable posterior approximation (Sec. 3). The authors claim this integration yields improved accuracy and interpretability, with both marginal and joint coverage guarantees (Sec. 3, last paragraph). Empirically, they evaluate a GRU-based classifier with a conformal head on MNIST (Sec. 4), reporting improvements in a "Conformal Prediction Error Rate" across three runs (Table 1; Sec. 5).

 **Strengths**

- Conceptually interesting integration of Bayesian quadrature with conformal prediction to probabilistically model nonconformity scores (Abstract; Sec. 1; Eq. (2)).
- Attention to hierarchical Bayesian modeling for dependencies across time/space (Eqs. (3)-(4)).
- Awareness of computational concerns and use of HMC/VI for scalability (Sec. 3).
- Clear statement of intended goal—robust UQ with distribution-free guarantees in high-stakes, spatio-temporal settings (Abstract; Sec. 1).

 **Weaknesses**

- Severe mismatch between scope and evaluation: the method is positioned for spatio-temporal, high-stakes domains, yet only MNIST classification is used (Abstract; Sec. 4), with no spatio-temporal dataset or task. This undermines claims of applicability to the target domain.
- No quantitative baselines against standard UQ/CP methods for classification or spatio-temporal CP; the paper acknowledges that "specific baseline results are not explicitly detailed" (Sec. 5).
- Insufficient statistical rigor: error rates are reported for 3 runs without confidence intervals/standard deviations, hypothesis tests, or calibration curves (Sec. 5).
- Methodological clarity gaps: (a) Eq. (5) does not describe HMC (it resembles a Langevin/SGLD step rather than leapfrog with momentum); (b) the algorithmic role of BQ is not specified (what integrals are computed, which kernels, how BQ modifies nonconformity and preserves CP validity); (c) the CP interface for classification is conflated with interval coverage (Eqs. (1), (6)), while for classification conformal prediction typically yields label sets, not intervals; (d) claims of joint coverage are made without a theorem or empirical evidence tailored to time-series/spatio-temporal dependence.
- Evaluation metrics are non-standard or undefined: "Bayesian Quadrature Estimated Coverage" is introduced but not formally defined or validated (Sec. 4).
- Analysis is largely descriptive; there is no ablation isolating BQ vs. hierarchical Bayes vs. HMC/VI contributions, no qualitative failure analyses, and no examples of prediction sets/intervals (Sec. 5).
- Reproducibility is limited: no pseudo-code or algorithm box; no description of calibration split protocol for CP (train/calibrate/test), no kernel/prior choices for BQ, no code.
- Some claims and references appear overly broad relative to the concrete method tested (e.g., asserting "both marginal and joint coverage guarantees" in Sec. 3 without proof or spatio-temporal evaluation).

? Questions

- BQ mechanism: What exactly is the integrand and measure in your Bayesian quadrature? How do you construct the GP prior (kernel choice, hyperparameters) over the nonconformity score function, and how is this used during conformal calibration/inference? Please provide precise algorithms/pseudo-code.
- Conformal validity: How does treating nonconformity scores as random via BQ interact with CP's distribution-free guarantees? Under what conditions are marginal coverage guarantees preserved? Please state a theorem or provide diagnostic coverage-vs-alpha curves across multiple seeds.
- Joint coverage: You claim both marginal and joint coverage (Sec. 3). What notion of joint coverage is targeted (e.g., pathwise, spatial field coverage)? Provide a statement and proof or targeted experiments on dependent data.
- HMC details: Eq. (5) does not describe HMC (which uses momenta and leapfrog integration). Are you in fact using Langevin/SGLD? If so, please correct the text and detail step sizes, accept/reject, and convergence diagnostics.
- Classification vs. intervals: For MNIST, conformal prediction typically yields label sets. How are "prediction intervals" defined for a 10-class classification task (Eqs. (1), (6))? If you mean set sizes or miscoverage rates, please formalize the metric and protocol.
- Calibration protocol: How do you split data for conformal calibration (train/calibration/test)? What is alpha? Please report empirical coverage at multiple alpha levels and the corresponding set sizes.
- Baselines: Please compare against standard CP baselines for classification (e.g., softmax-score CP, APS/APS+, CQR-like surrogates for classification), and for time series/spatio-temporal CP (e.g., STACI, weighted/online CP variants), as well as Bayesian deep ensembles or GP-based baselines where applicable.
- Spatio-temporal evaluation: Why is there no spatio-temporal dataset? At minimum, evaluate on a bona fide temporal or spatio-temporal benchmark (e.g., traffic, weather reanalysis, human activity time series, wildfire forecasting) to substantiate the claimed domain of application.
- Metric definition: What is "Bayesian Quadrature Estimated Coverage"? Please define it formally, justify its use, and relate it to standard CP diagnostics.
- Ablations and cost: Provide ablations removing BQ or hierarchical Bayes to quantify their contributions. Report computational overhead (wall-clock, FLOPs) versus baselines, and convergence diagnostics for inference.

! Limitations

- Domain mismatch: Current evaluation on MNIST does not establish performance in spatio-temporal, high-stakes domains claimed as the target.
- Reliance on Gaussian assumptions for nonconformity score modeling (Eq. (2)) may conflict with the distribution-free spirit of CP and affect validity if used improperly in calibration.

- Exchangeability assumptions may be violated in spatio-temporal settings; additional techniques (e.g., weighted/online CP, block/cluster CP) or theoretical adjustments are needed for dependence.
- Computational overhead of Bayesian inference could be significant for real-time applications; careful profiling and approximate inference strategies are required.
- Potential negative societal impact: Overstating reliability of UQ in high-stakes domains without domain-appropriate validation could induce misplaced trust and harmful decisions.

Image Evaluation

Cross-Modal Consistency: 18/50

Textual Logical Soundness: 10/30

Visual Aesthetics & Clarity: 14/20

Overall Score: 42/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Major 1: Core task is spatio-temporal UQ, but experiments are on static MNIST digits (domain mismatch). Evidence: Sec 4 “We conducted experiments using the MNIST dataset”
- Major 2: Reported “sanity check” parameter count conflicts with stated architecture; a 28→64 GRU alone exceeds 9,420 params. Evidence: Sec 4 “confirmed the total parameter count matched the expected 9,420”
- Major 3: One of two evaluation metrics (Bayesian Quadrature Estimated Coverage) is promised but never reported in Results. Evidence: Sec 4 “Evaluation Metrics... and the Bayesian Quadrature Estimated Coverage”; Sec 5 Table 1 only shows CP Error Rate
- Minor 1: Table 1 lacks the target α (coverage level) and definition of “Conformal Prediction Error Rate,” hindering interpretability.
- Minor 2: Method claims hierarchical Bayesian modeling, but the implementation section describes only a single-layer GRU with a linear head; no concrete hierarchical components are specified or validated.

2. Text Logic

- Major 1: HMC update is incorrect (closer to Langevin/MALA than HMC), undermining methodological credibility. Evidence: Sec 3, Eq.(5) “ $\theta^{i+1} = \theta^i + \epsilon \nabla \log P(\theta^i) + N(0, \epsilon^2)$ ”
- Major 2: Strong claim of joint coverage guarantees lacks theorem, proof, or empirical verification. Evidence: Sec 3 “offering both marginal and joint coverage guarantees”
- Minor 1: “Bayesian quadrature” use is not formalized (no GP prior on integrand, kernel, or integral objective); modeling $s(x,y) \sim N(\mu, \sigma^2)$ is not BQ per se. Evidence: Sec 3, Eq.(2) “ $s(x_t, y_t) \sim N(\mu_t, \sigma_t^2)$ ”
- Minor 2: No baselines or statistical tests; improvements across runs could reflect randomness.

3. Figure Quality

- Minor 1: Table 1 is legible but fails the Figure-Alone test: lacks α , dataset splits context, and baseline comparison in the caption.
- Minor 2: No architecture diagram; given emphasis on hierarchical Bayesian modeling and BQ-CP integration, a block diagram with module names and data-flow would clarify the method.

Key strengths:

- Timely objective: combining conformal guarantees with Bayesian richness for UQ.
- Clear motivation and practical interest in high-stakes domains.
- Simple baseline architecture could ease adoption if method were specified rigorously.

Key weaknesses:

- Fundamental domain mismatch (spatio-temporal claim vs MNIST digits).
- Methodological inaccuracies (HMC equation), and vague BQ integration.
- Unsupported flagship claims (joint coverage), missing promised metric, and parameter count inconsistency.
- Lack of baselines, ablations, or statistical rigor in results.

Recommendations: Align experiments with genuine spatio-temporal data, correct HMC derivation, provide a precise BQ formulation and algorithm, include baselines with CIs, report both metrics, and add an architecture/workflow figure with clear labels and symbols.

 Scores

Originality: 3

Quality: 1

Clarity: 2

Significance: 2

Soundness: 1

Presentation: 2

Contribution: 2

Rating: 3

AI Review from SafeReviewer

 AI Review from SafeReviewer will be automatically processed

 Summary

This paper presents a novel approach to uncertainty quantification (UQ) in spatio-temporal models, integrating Bayesian and conformal prediction methods. The authors propose a framework that combines the probabilistic richness of Bayesian quadrature with the distribution-free guarantees of conformal prediction. The method is evaluated on the MNIST dataset, demonstrating improvements in prediction accuracy and reliability.

 Suggestions

To improve the paper, I suggest focusing on three main areas: theoretical grounding, experimental validation, and clarity of presentation. Firstly, regarding the theoretical aspects, it is crucial to provide a rigorous definition of the non-conformity score and the specific conformal prediction method employed. This should include a clear mathematical formulation and a discussion of its properties, particularly concerning joint coverage. The authors should either provide a theoretical analysis demonstrating the joint coverage guarantees of their method or explicitly acknowledge the limitations if such guarantees cannot be provided. Additionally, a more detailed explanation of how Bayesian quadrature is integrated with conformal prediction is needed. This could involve a step-by-step breakdown of the algorithm or a pseudocode representation to clarify the interaction between the two components. Furthermore, the authors should address the handling of missing data, which is a common challenge in spatio-temporal datasets. They should discuss how their method accounts for missing values and provide empirical evidence to support their claims.

Secondly, the experimental validation needs significant enhancement. The current evaluation on the MNIST dataset is insufficient to demonstrate the effectiveness of the proposed method for spatio-temporal problems. The authors should include experiments on at least one real-world spatio-temporal dataset relevant to the target applications, such as climate science or epidemiology. This would provide a more convincing demonstration of the method's practical utility. Moreover, a

thorough comparison with state-of-the-art methods in spatio-temporal uncertainty quantification is essential. The authors should select appropriate baselines and report performance metrics that allow for a fair comparison. This would help to establish the significance of the proposed method's performance improvements. Finally, the computational cost of the proposed method should be empirically evaluated and compared with existing approaches. This would provide valuable insights into the practical feasibility of the method for large-scale spatio-temporal datasets.

Lastly, the clarity of presentation can be improved by addressing the grammatical errors and awkward phrasing throughout the paper. A careful proofreading is necessary to ensure that the writing is clear and concise. Additionally, the authors should cite relevant works in the field of conformal prediction and spatio-temporal modeling to provide a comprehensive overview of the related research. This would help to contextualize their contribution and highlight the novelty of their approach. By addressing these suggestions, the authors can significantly strengthen the paper and make a more compelling case for their proposed method.

? Questions

1. Could the authors provide a clear definition of the non-conformity score and the conformal prediction method used in the paper? This will help in understanding the theoretical underpinnings of the proposed approach.
2. Can the authors provide a theoretical guarantee for the joint coverage of the conformal prediction intervals? If not, can they discuss the limitations of their approach in this regard?
3. Could the authors provide a more detailed explanation of how the Bayesian quadrature method is integrated with conformal prediction? A step-by-step description or a pseudocode would be beneficial.
4. Why was the MNIST dataset chosen for evaluation? Can the authors evaluate their method on at least one real-world spatio-temporal dataset to demonstrate its effectiveness in the target domain?
5. Can the authors compare their method with state-of-the-art methods in spatio-temporal uncertainty quantification? This would help in assessing the significance of the proposed method's performance improvements.
6. Could the authors provide a comparison of the computational cost of the proposed method with existing methods? This would help in understanding the practical feasibility of the method.
7. What are the limitations of the proposed method? Can the authors discuss the potential scenarios where their approach might not perform well?
8. Could the authors clarify how their method handles missing data in spatio-temporal datasets?

📊 Scores

Soundness: 1.75

Presentation: 2

Contribution: 2

Confidence: 4

Rating: 3

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights